

RSP-ADMM-PID: Robust Safety-Projected Adaptive Glucose Control Across Heterogeneous Virtual Cohorts

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Abstract— Automated insulin delivery must regularize changes in blood sugar resulted from meals, activity, sensor error and patient variability without causing hyperglycemia or hypoglycemia. In order to do so, we propose Robust Scenario-Predicted, safety-projected Alternating Direction Method of Multipliers PID (RSP-ADMM-PID), an online gain adapting method that combines a Continuous glucose monitoring (CGM) driven observer, short horizon uncertainty scenarios, three step ADMM updates and a predictive safety projection. A 24-h benchmark used an extended Bergman model with meals, exercise and circadian effects, CGM noise and lognormal inter subject variation. From the benchmark, thirty held-out normal, type-1 diabetes (T1D) and insulin-treated type-2 diabetes (T2D) virtual subjects were evaluated with matched disturbances against PID, successive linearization LQG, Proximal Policy Optimization (PPO) and PPO-PID. RSP-ADMM-PID over achieved the highest Blood Glucose Time in 70 $\frac{mg}{dL}$ to 180 $\frac{mg}{dL}$ range percentage of $87.16 \pm 5.31\%$ for T1D and $89.85 \pm 4.79\%$ for T2D, over the thirty simulated patients. Additionally, Blood Glucose Time Below Range (TBR) of 70 $\frac{mg}{dL}$ was 0% in both cohorts for the proposed method. Moreover, RSP-ADMM-PID also measured the lowest RMSE over 24-h to matched normal physiology at $23.76 \pm 8.71 \frac{mg}{dL}$ and $17.50 \pm 3.61 \frac{mg}{dL}$ for T1D and T2D respectively with paired t-test ($p \leq 2.36 \times 10^{-5}$).

Keywords— Bergman model, ADMM, adaptive PID, robust control, safety filtering

I. INTRODUCTION

Continuous glucose monitoring (CGM), an insulin pump and a feedback algorithm, form an automated insulin-delivery (AID) system [1]. International consensus defines 70 – 180 $\frac{mg}{dL}$ as the principal glucose target range for most nonpregnant people with T1D or T2D and recommends reporting time below and above range rather than a mean alone [2]. Clinical closed-loop studies have demonstrated the feasibility of automated dosing in T1D and, more recently, in insulin-treated T2D [3]. Nevertheless, delay in insulin secretion, unannounced meals, physical activity, CGM error

and changing insulin sensitivity create a nonlinear, partially observed safety-critical control problems [4].

Related works. Model-based evaluation began with the Bergman minimal model; meal and population models later supported reproducible in-silico pre-clinical studies [5]. Literature showed PID is a computationally attractive controller, that is clinically established, whereas, model-predictive controller (MPC) can handle prediction and constraints explicitly [6]. Limitations are complementary of these two families of controllers: Fixed PID gains cannot anticipate uncertainty, whereas an accurate MPC can be computationally demanding [6]. Auxiliary safety feedback has therefore become a design layer incorporated in controller design to make use of MPC advantage of constraints, while keeping the computational attractiveness of PID controller [7].

Reinforcement learning (RL) has been demonstrated to have the ability to learn nonlinear policies from randomized simulation for many applications [8,9]. As a result, PPO and other deep-RL variants have been evaluated for basal and fully closed-loop glucose control [10]. Yet RL in healthcare requires explicit scrutiny of distribution shift, leakage validation, interpretability and action safety [10]. Recent safe-RL researches support a hybrid position: learning or optimization should be paired with a separately auditable safety mechanism [11].

Gap. Despite these advances, a methodological gap persists between the agility of learning-based agents and the transparency of clinical control stands. While RL offers a path toward personalized control, its reliance on heavy, black-box architectures complicates the safety verification required for life-critical medical devices [12]. Conversely, PID loops and standard MPC formulations often struggle to maintain stability in real-world stochastic disturbances [13]. Moreover, existing hybrid architectures typically impose solving complex non-convex optimization problems that compromises glycemic efficiency [14]. This becomes a critical gap for a control architecture, integrating predictive robustness of scenario-based optimization with the interpretability of PID.

Contributions. To cover the existing gaps, we suggested the following novel procedure:

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² Control Benchmarks

³ Method Benchmarks

⁴ RL Benchmarks

- i. A benchmark with multiple stochastic meals, activity, sensor noise and patient uncertainty,
- ii. RSP-ADMM-PID, which performs online scenario-aware PID-gain updates using ADMM and projects the insulin command onto a predicted-safe interval,
- iii. Comparison of RSP-ADMM-PID against five controller families, including state-of-art reinforcement learning methods in PID-tuning,
- iv. An ablation study isolating the individual impacts of the robust scenario predictions and the safety projection, demonstrating Time in Range (TIR) improvements, while strictly eliminating hypoglycemia.

Organization. The rest of the paper is organized as follows. Section II introduces the Bergman physiological model, detailing the incorporation of meal, exercise and circadian disturbances. Section III formulates the proposed RSP-ADMM-PID control architecture, the scenario robust ADMM optimization for online PID-gain adaptation and the predictive safety projection layer. Section IV outlines the benchmark setup and introduces the baseline controllers, including SL-LQG [15] and reinforcement learning-based approaches. Section V presents comparative simulation results, analyzing clinical metrics such as Time in Range (TIR) and Time Below Range (TBR). Finally, Section VI concludes the paper and highlights potential directions for future research.

II. SAFE GLOUCSE CONTROL PROBLEM

This section presents the robust glucose control problem under physiological uncertainties and actuator constraints. We first define an extended Bergman-type state-space model capturing the core glucose and insulin dynamics alongside meal absorption, time-varying insulin sensitivity, and CGM sensor noise. Subsequently, we formalize the control objective.

A. Extended physiological modeling and virtual cohort

Let the simulation time be t in minutes. For a subject, define the physiological state

$$x(t) = [G(t) \ I(t) \ X(t) \ B(t) \ G_s(t)]^T.$$

and observable state as

$$y(t) = [B(t) \ G_s(t)]^T.$$

Here G is plasma glucose concentration, I is plasma insulin concentration, X is the remote insulin-effect state, B is insulin-on-board (IOB) and G_s is the lagged sensor state. The manipulated input is the commanded insulin rate $u(t)$.

The extended Bergman-type model, inspired by [16] is modeled as:

$$\dot{G} = -p_1(G - G_b) - \alpha(t)XG + R_a(t) + R_\beta(G, I). \quad (1a)$$

$$\dot{I} = -p_2(I - I_b) + k_u u(t) + k_\beta S_\beta(G). \quad (1b)$$

$$\dot{X} = -p_3 X + (I - I_b). \quad (1c)$$

$$\dot{B} = -\frac{1}{\tau_B} B + \frac{1}{\tau_B} u(t). \quad (1d)$$

$$\dot{G}_s = \frac{1}{\tau_s} (G - G_s). \quad (1e)$$

In (1), G_b and I_b denote basal glucose and insulin concentrations; p_1, p_2 and p_3 are positive model parameters; k_u converts pump delivery to plasma insulin input; k_β scales endogenous secretion and $S_\beta(G)$ represents the phenotype-specific endogenous secretion law. Numerical values of parameters of system in (1) and how they affect type of diabetes is given in Table 1.

The insulin-sensitivity modulation can be factorized [17] as:

$$\alpha(t) = \alpha_c(t) \alpha_e(t). \quad (2)$$

and given that a maximum and minimum on exists insulin-sensitivity modulation, we can state that:

$$0 < \alpha_{\min} \leq \alpha(t) \leq \alpha_{\max} < \infty \quad (3)$$

For meal m , let C_m denote carbohydrate amount, t_m the meal occurrence time, and τ_m its absorption constant. The meal appearance rate is modeled, inspired by [16] as:

$$R_a(t) = \sum_m \frac{C_m}{V_G} \frac{t - t_m}{\tau_m^2} e^{-(t-t_m)/\tau_m} \mathbf{1}_{\{t \geq t_m\}}. \quad (4)$$

where V_G is the glucose distribution volume and

$$\mathbf{1}_{t \geq t_m} \triangleq \begin{cases} 1. & t \leq t_m \\ 0. & \text{o.w.} \end{cases}$$

Additionally, CGM measurement model is:

$$y_k = \max(\min(G_s(t_k) + b_s + \sigma_s \varepsilon_k, y_{\min}), y_{\max}) \quad (5)$$

Where $y_{\min} = 40$ and $y_{\max} = 400$ based on [18].

The correlated measurement error process is then assumed, similar to work in [19], to be:

$$\begin{aligned} \varepsilon_k &= \phi_\varepsilon \varepsilon_{k-1} + \sqrt{1 - \phi_\varepsilon^2} \eta_k. \\ \eta_k &\sim \mathcal{N}(0,1). \quad |\phi_\varepsilon| < 1. \end{aligned} \quad (6)$$

What is more, due to pump limitations [20], the physical actuator constraint is:

$$0 \leq u_k \leq u_{\max}. \quad (7)$$

Finally, we note that a designed controller does not have direct access to $G(t)$, $R_a(t)$, $I(t)$, $X(t)$ or the exercise state.

B. Problem Statement

The problem in this section, would be to find a $u(t)$ for a set horizon, such that given modeling of (1), in presence of (2), (4), (6) and (7), a safe and similar to a non-diabetic trend in behavior of (1) is seen, especially in terms of blood glucose level in a metric such as RMSE. What is more, the performance of a method is evaluated as good, given that a metric such as TIR is closer to 100 percent, while metrics such as TBR and Time Above Range (TAR) and the amount of given insulin are lower.

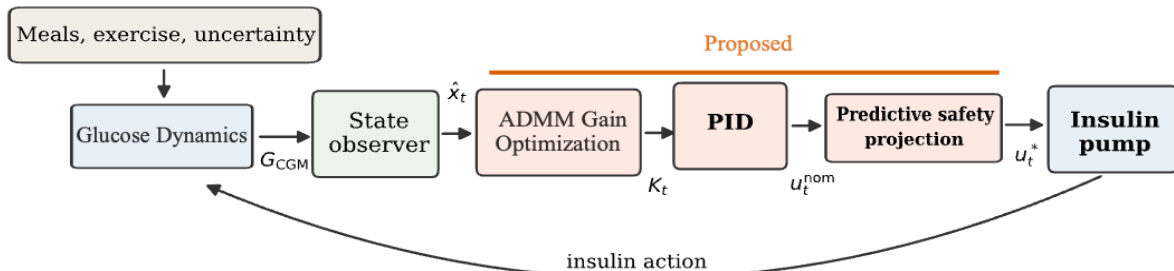


Figure 1. Proposed RSP-ADMM-PID controller. The virtual cohort receives meals, exercise, physiological uncertainty and CGM disturbances. A Kalman filter estimates the control-relevant state from CGM measurements. Scenario-based ADMM updates the PID gains, after which the adaptive PID generates the nominal insulin pump command input. A predictive safety projection then limits the insulin pump command before it is applied to the insulin pump. The resulting administered insulin action closes the feedback loop.

Table 1. Summary of parameters for glucose-insulin state-space equations. Positive patient parameters are independently lognormal with the stated coefficient of variation (CV); truncations are shown explicitly. Distributions are declared robustness-test choices anchored to control-relevant model.

Parameter (unit)	Normal	T1D	T2D	Randomization
$G_b(\text{mg/dL});$ $I_b(\text{mU/L})$	100; 15	110; 15	110; 20	$G_b + N(0, 3^2); I_b \text{ CV12\%}$
$p_1;$ $p_2(\text{min}^{-1})$	0.028; 0.025	0.012; 0.025	0.018; 0.025	CV22; CV18%
$p_3((\text{mU}/\text{L})^{-1}\text{min}^{-2});$ $n(\text{min}^{-1})$	$1.30 \times 10^{-5};$ 0.0926	Same as Normal	$0.806 \times 10^{-5};$ 0.0926	CV28%; 15%
$k_\beta; k_u$	0.060; 0.90	0; 0.90	0.020; 0.85	CV30% if nonzero; 18%
$u_b; u_{\max}(\text{U/h});$ $\tau_b(\text{min})$	0; 0; 60	0.90; 5; 60	0.45; 4; 70	basal CV12%; τ_b CV18%
$\tau_s(\text{min}); b_s;$ $\sigma_s(\text{mg/dL})$	10; 0; 4	10; 0; 5	10; 0; 5	τ_s CV20% 5,18; $b_s \sim N(0, 3^2); \sigma_s$ CV15% [2.5,8]
Meals	breakfast/lunch/dinner medians 48/68/74 g; optional snack 24 g (prob. 0.60)			times truncated normal; carbs CV25%; τ_m CV22% [25,80] min
Activity; numerical grid	Prob. 0.78; start 17.2 ± 0.83 h [14.5,20.5]; 30 – 75 min; intensity 0.35 – 0.85			RK4 1min; CGM/control 5min; 24h

III. RSP-ADMM-PID METHOD

In the current work, we will be using a step-adaptive PID controller, where the PID controller is tasked with controlling the input in (1) through observing the error. What is more, the gains are selected through optimizing a defined metric, where such optimization is parametrized based on estimating the states of (1) using a Kalman filter and solved through ADMM. Figure 1 presents a graphical abstract of the methodology presented here.

A. Gain-adaptive PID

Following the classical literature in, one would define the glucose tracking error at each time step k :

$$e_k = y_k - G_{\text{ref}}. \quad (8)$$

Where $G_{\text{ref}} = 110 \frac{\text{mg}}{\text{dl}}$ is an optimal glucose level. What is more, integral state can be defined as:

$$\xi_k = \min(\max(\xi_{k-1} + \Delta t e_k, -\xi_{\max}), \xi_{\max}). \quad (9)$$

where $\xi_{\max}$ is a positive hyperparameter. Additionally, filtered derivative signal at each step, given the (19) can be written as:

$$d_k = \lambda_d d_{k-1} + (1 - \lambda_d) \frac{e_k - e_{k-1}}{\Delta t}, 0 \leq \lambda_d < 1 \quad (10)$$

Let us defined the normalized adaptive gain vector to be:

$$\theta_k = [\theta_{p,k} \quad \theta_{i,k} \quad \theta_{d,k}]^T \in \mathbb{R}^3 \quad (11)$$

The PID gains can then be defined as the following as each step:

$$K_{p,k} = K_p^{\min} + \theta_{p,k}(K_p^{\max} - K_p^{\min}). \quad (12a)$$

$$K_{i,k} = K_i^{\min} + \theta_{i,k}(K_i^{\max} - K_i^{\min}) \quad (12b)$$

$$K_{d,k} = K_d^{\min} + \theta_{d,k}(K_d^{\max} - K_d^{\min}) \quad (12c)$$

The unconstrained PID controller is then:

$$u_k^{\text{raw}} = u_b + K_{p,k} e_k + K_{i,k} \xi_k + K_{d,k} d_k \quad (13)$$

The constrained nominal control input can then be proposed as a saturation to feasible pump inputs:

$$u_k^{\text{nom}} = \min(\max(u_k^{\text{raw}}, 0), u_{\max}) \quad (14)$$

Moreover, the integral state wind up due to saturation in (14) is prevented trough integral term being set to zero at the u_k^{nom} limits.

B. State estimation and robust optimization goal

Let $x_k = x(t_k)$ presents the hidden states vector in (1) and $y_k = y(t_k)$ present all the states observable states in (1). The nonlinear state-space model can be then defined as:

$$x_{k+1} = f(x_k, u_k, \omega_k). \quad (15a)$$

$$y_k = h(x_k) + v_k. \quad (15b)$$

In order to estimate current glucose and IOB, the extended Kalman filter prediction step can be used as the following, based on [21]:

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1}, 0). \quad (16a)$$

$$P_{k|k-1} = A_k P_{k-1|k-1} A_k^T + Q. \quad (16b)$$

Where matrix Q is a fixed hyperparameter and

$$A_k = \frac{\partial f}{\partial x} \Big|_{\hat{x}_{k-1|k-1}, u_{k-1}}. \quad (17)$$

The measurement update is then:

$$u_k = y_k - h(\hat{x}_{k|k-1}) \quad (18a)$$

$$H_k = \frac{\partial h}{\partial x} \Big|_{\hat{x}_{k|k-1}} \quad (18b)$$

$$S_k = H_k P_{k|k-1} H_k^T + R \quad (18c)$$

$$K_k = P_{k|k-1} H_k^T S_k^{-1} \quad (18d)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k v_k \quad (18e)$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \quad (18f)$$

The IOB and Glucose state then can be reconstructed from (18), given that type of diabetes is known.

C. Scenario-based robust optimization objective

The gain vector is selected by solving the finite-horizon optimization problem using ADMM over the horizon H_o where we intend to optimize the gain . The three uncertainty scenarios are $\mathcal{S} = \{1,2,3\}$. For each controller update, three uncertainty scenarios are generated from the same estimated patient state. Scenario 1 assumes elevated insulin sensitivity with no additional hidden meal

appearance; Scenario 2 is the nominal prediction; and Scenario 3 assumes reduced insulin sensitivity together with increased hidden meal appearance.

For a candidate gain vector, let $\hat{G}$ at future step $k + j$ under scenario s denote predicted glucose per (13). We define three scalar loss functions as

$$r(G) = G - G_{\text{ref}} \quad (19a)$$

$$r_L(G) = \max(0, G_L - G). \quad (19b)$$

$$r_H(G) = \max(0, G - G_H). \quad (19c)$$

where $G_L = 70$ mg/dL and $G_H = 180$ mg/dL are the maximum and minimum safe ranges [2]. What is more, we numerically set $G_{\text{ref}} = 110$ mg/dL as a nominal in range reference.

We define the cost at k as a function of the gain vector as the following:

$$J_s(\theta) = \frac{1}{H_o} \sum_{j=1}^{H_o} \left[w_e r \left(\hat{G}_{k+j}^{(s)}(\theta) \right)^2 + w_L r_L \left(\hat{G}_{k+j}^{(s)}(\theta) \right)^2 + w_H r_H \left(\hat{G}_{k+j}^{(s)}(\theta) \right)^2 + w_u \left(\hat{u}_{k+j}^{(s)}(\theta) \right)^2 \right] \quad (20)$$

The robust objective is then simply the worst case over three parameter cases:

$$\hat{f}(\theta) = \max_{s \in \mathcal{S}} J_s(\theta). \quad (21)$$

The gain objective with temporal and reference regularization for more controlled optimization behavior is then defined as:

$$F(\theta) = \hat{f}(\theta) + \frac{\lambda_s}{2} \|\theta - \theta_{k-1}\|_2^2 + \frac{\lambda_0}{2} \|\theta - \theta_0\|_2^2 \quad (22)$$

Introduce the splitting variable $z = \theta$ for gains during horizon H_o and then the constrained problem to solve becomes:

$$\begin{aligned} \min_{\theta, z} \quad & \hat{f}(\theta) + \frac{\lambda_s}{2} \|\theta - \theta_{k-1}\|_2^2 + \frac{\lambda_0}{2} \|\theta - \theta_0\|_2^2 + \iota_\theta(z). \\ \text{s.t} \quad & \theta - z = 0. \end{aligned} \quad (23)$$

Where the additional feasibility constraint

$$\iota_\theta(z) \triangleq \begin{cases} 0, & z \in \Theta. \\ +\infty, & z \notin \Theta \end{cases} \quad (24)$$

D. ADMM gain update

The gain subproblem is solved using a three-iteration linearized ADMM scheme. The auxiliary variable and dual updates are exact, whereas the primal gain update uses a proximal first-order linearization.

The scaled augmented Lagrangian is:

$$\begin{aligned} \mathcal{L}_{\rho_{\text{AD}}}(\theta, z, d) = & \hat{f}(\theta) + \frac{\lambda_s}{2} \|\theta - \theta_{k-1}\|_2^2 + \\ & \frac{\lambda_0}{2} \|\theta - \theta_0\|_2^2 + \iota_\theta(z) + \\ & \frac{\rho_{\text{AD}}}{2} \|\theta - z + d\|_2^2 - \frac{\rho_{\text{AD}}}{2} \|d\|_2^2 \end{aligned} \quad (25)$$

where $d \in \mathbb{R}_+$ is the dual variable for (25) and ρ_{AD} is the augmented Lagrangian hyper-parameter. We then choose a sub-gradient g^ℓ over the subdifferential set $\partial J_s(\theta)$ given multiple scenarios,

$$g^\ell \in \{\partial J_s(\theta) : s \in \mathcal{A}^\ell\} \quad (26)$$

where:

$$\mathcal{A}^\ell = \arg\max_s J_s(\theta^\ell) \quad (27)$$

With the definition of q^ℓ :

$$q^\ell = g^\ell + \lambda_s(\theta^\ell - \theta_{k-1}) + \lambda_0(\theta^\ell - \theta_0) \quad (28)$$

the linearized primal update can be written as:

$$\theta^{\ell+1} = \underset{\theta}{\operatorname{argmin}} \left\{ \langle q^\ell, \theta - \theta^\ell \rangle + \frac{1}{2\eta} \|\theta - \theta^\ell\|_2^2 + \frac{\rho_{\text{AD}}}{2} \|\theta - z^\ell + d^\ell\|_2^2 \right\} \quad (29)$$

Where $\eta^{(-1)}$ is proportional to Learning rate. Its closed-form solution then becomes:

$$\theta^{\ell+1} = \frac{\theta^\ell - \eta q^\ell + \eta \rho_{\text{AD}}(z^\ell - d^\ell)}{1 + \eta \rho_{\text{AD}}} \quad (30)$$

What is more, the trust region is defined on the simplex Δ_θ :

$$\mathcal{T}_k = \{\theta : \|\theta - \theta_{k-1}\|_\infty \leq \Delta_\theta\} \quad (31)$$

Which is set to be a box constraint [0.1]. The primal update is projected as:

$$\theta^{\ell+1} \leftarrow \Pi_{\mathcal{T}_k}(\theta^{\ell+1}) \quad (32)$$

Where the projection is defined via second norm:

$$\Pi_{\mathcal{T}_k}(\theta^{\ell+1}) = \underset{\psi}{\operatorname{argmin}} \|\theta^{\ell+1} - \psi\|_2^2 \text{ s.t } \psi \in \mathcal{T}_k. \quad (33)$$

The auxiliary-variable update in ADMM is then written as:

$$z^{\ell+1} = \underset{z}{\operatorname{argmin}} \frac{1}{2} \rho_{\text{AD}} \|\theta^{\ell+1} - z + d^\ell\|_2^2 \quad (34)$$

Hence $z^{\ell+1} = \Pi_\Theta(\theta^{\ell+1} + d^\ell)$. Finally, the dual update simply becomes:

$$d^{\ell+1} = d^\ell + \theta^{\ell+1} - z^{\ell+1} \quad (35)$$

Given the initialization is $z^0 = \Pi_\Theta(\theta_{k-1})$, $d^0 = 0$.

E. Predictive safety projection

Let $H_s > H_o$ be the safety prediction horizon. For a candidate constant control input v , define the safety-feasible set as:

$$\mathcal{U}_k = \left\{ v : 0 \leq v \leq \min(u_k^{\text{nom}}, u_{\text{max}}), \min_{1 \leq j \leq H_s} \hat{G}_{k+j}(v) \geq G_{\text{safe}} \right\}. \quad (36)$$

where we propose a safe glucose level above G_L as $G_{\text{safe}} = 75 \frac{\text{mg}}{\text{dL}}$. The projected control input at time step k is then:

$$u_k = \begin{cases} \underset{v \in \mathcal{U}_k}{\operatorname{argmin}} (v - u_k^{\text{nom}})^2, & \mathcal{U}_k \neq \emptyset. \\ 0, & \mathcal{U}_k = \emptyset. \end{cases} \quad (37)$$

IV. EXPERIMENTAL PROTOCOL¹

We simulated thirty test subjects per cohort and diabetic type using seeds $s_i \propto 17i$ and day seeds $d_i \propto 31i$, $i = 0, \dots, 29$, with initializations in [8]. Every method received the same subject, day and noise realization; matched normal physiology used the same s_i and d_i . Tuning selection used disjoint seeds. The fixed PID through Ziegler–Nichols tuning [22] used $K = [0.028, 0.00012, 0.045]^T$ for T1D and $K = [0.024, 0.00010, 0.040]^T$ for T2D. Successive-linearization LQG (SL-LQG) [15], recomputed a discrete LQR about each EKF estimate. Non-robust ADMM-PID used nominal scenario and no projection.

Direct PPO mapped a ten-dimensional observation to normalized pump rate. PPO-PID mapped it to the three normalized gains. Both used a 32-unit actor, Gaussian latent action followed by a sigmoid, clipped with $\varepsilon = 0.2$, GAE ($\gamma = 0.995, \lambda = 0.95$), sixty epochs per update and Adam optimizer [23]. Each method was trained for 24 updates with six randomized T1D/T2D days per update and three independent seeds (11, 29, 47) for policies. The best checkpoint per seed used a fixed validation set. Seed selection

¹ Implementation details can be found at the [GitHub](#) Repository.

used another thirty-subject-per-cohort validation set. Test seeds were never queried to prevent data leakage.

For each cohort endpoint, the proposed method versus each baseline was tested with a t-test to measure statistical significance [24].

V. RESULTS

This section evaluates RSP-ADMM-PID across the held-out *T1D* and insulin-treated *T2D* simulated cohorts under the matched 24-h experimental protocol described in Section IV.

A. Primary and secondary outcomes

Table 2 reports all cohorts and methods comparison. Normal physiology achieved $96.81 \pm 3.22\%$ TIR without exogenous insulin. Moreover, Figure

In *T1D*, the proposed method increased TIR by 1.91% versus fixed PID, 2.18% versus PPO-PID and 2.56% against ADMM-PID. In *T2D*, the corresponding gains were 1.36%, 0.75% and 1.76%, respectively. Notably, TIR improvements were positive for RSP-ADMM-PID ($p \leq 9.75 \times 10^{-6}$ in *T1D*; $p \leq 2.36 \times 10^{-5}$ in *T2D*). Additionally, the RMSE was lower than all baselines ($p = 4.66 \times 10^{-9}$). Against the strongest baseline for this endpoint, the mean difference was -4.30 mg/dL (95% bootstrap: $[-5.10, -3.56]$) in *T1D* versus fixed PID and -1.69 mg/dL ($[-1.94, -1.46]$) in *T2D* versus PPO-PID. TBR70 was 0% in all cohorts for RSP-ADMM-PID, PPO and LQG only. In addition, no method produced time below 54 mg/dL. We note that *T2D* and *T1D* performance for our proposed method required more mean daily insulin than fixed PID or PPO-PID, so the advantage is mainly glycemic, not dose-minimizing.

Runtime was 4.04 ± 0.13 s per 24-h *T1D* simulation and 3.91 ± 0.06 s for *T2D* on the evaluation host, which was similar to that of inference on PPO, PPO-PID and SL-

LQG. We should note that although Fix-gain PID is worse in terms of most metrics, it dominates in terms of computational effort.

B. Performance comparison

Figure 2 shows that the proposed methods mean response remains closest to matched normal after meals without visible low-glucose overshoot. In-contrast Direct PPO underperformed in *T1D* despite improving validation return, illustrating why a mechanistic safety layer matter.

Robust scenarios method produces larger, structured post-meal changes than the nominal ADMM alone. In *T1D* the safety layer intervened $19.5 \pm 17.6 \frac{\text{times}}{\text{day}}$. However, most interventions were small rate reductions and not low-glucose events. It was inactive in the *T2D* test cohort because its risk gate did not trigger, showing even better performance.

C. Robustness and method performance gain analysis

Figure 3 showcases per-subject outcome distributions. As seen from Figure 3, all methods using a PID as the controller were more robust to change in subjects. In addition, as seen in figure 3, RSP and PID were both had the smallest scatter among other methods. The superiority of RSP then can be established, given that fixed PID is worst in the most metrics in table 2.

In addition, Table 3 demonstrates effect of each part of the RSP-ADMM-PID, using the first 15 held-out subjects as a pre-indexed ablation subset. In *T1D*, RSP-ADMM-PID improved TIR by 2.17% and RMSE110 by $5.28 \frac{\text{mg}}{\text{dL}}$ over nominal ADMM-PID, but retained 0.023% TBR. Safety filter removed TBR while leaving TIR nearly unchanged, showcasing minimal trade-off. In *T2D*, the safety gate never activated and therefore improvements came from the cost function design.

Table 2. Held-out 24-h performance (mean $\pm$ std, $n = 30$ virtual subjects/cohort). TIR, Time in Transition (TIT), TBR and TAR are shown in percentage; N-RMSE is RMSE to the matched normal trace ($\frac{\text{mg}}{\text{dL}}$) per day; insulin is $\frac{\text{U}}{\text{day}}$.

Cohort	Method	TIR 70–180	TIT 80–140	TBR < 70	TAR > 180	N-RMSE	Insulin
Normal	Normal physiology	96.81 ± 3.22	85.18 ± 6.27	0 ± 0	3.19 ± 3.22	0	0
T1D	Fixed PID	85.25 ± 5.26	69.99 ± 5.62	0.16 ± 0.67	14.59 ± 5.24	28.05 ± 7.45	49.54 ± 6.96
	SL-LQG	81.12 ± 7.54	63.17 ± 9.15	0 ± 0	18.88 ± 7.54	35.56 ± 8.56	38.10 ± 4.58
	Direct PPO	67.49 ± 9.72	51.64 ± 8.50	0 ± 0	32.51 ± 9.72	62.20 ± 14.41	34.74 ± 0.71
	PPO-PID	84.98 ± 5.22	68.58 ± 6.06	0.43 ± 1.33	14.59 ± 5.26	28.19 ± 7.29	50.13 ± 7.17
	ADMM-PID	84.60 ± 5.35	72.10 ± 6.56	0.01 ± 0.06	15.39 ± 5.35	29.22 ± 8.54	45.48 ± 6.34
	RSP-ADMM-PID	87.16 ± 5.31	75.14 ± 6.10	0 ± 0	12.84 ± 5.31	23.76 ± 8.71	49.63 ± 8.48
T2D	Fixed PID	88.49 ± 4.91	73.97 ± 6.25	0 ± 0	11.51 ± 4.91	20.17 ± 3.44	35.57 ± 7.24
	SL-LQG	84.64 ± 6.54	64.69 ± 8.90	0 ± 0	15.36 ± 6.54	29.27 ± 3.61	16.29 ± 2.59
	Direct PPO	84.69 ± 6.54	67.00 ± 8.21	0 ± 0	15.31 ± 6.54	28.45 ± 4.82	26.72 ± 0.87
	PPO-PID	89.10 ± 4.94	75.06 ± 5.66	0 ± 0	10.90 ± 4.94	19.19 ± 3.50	38.61 ± 8.41
	ADMM-PID	88.09 ± 5.04	73.83 ± 5.98	0 ± 0	11.91 ± 5.04	20.80 ± 3.62	32.72 ± 7.02
	RSP-ADMM-PID	89.85 ± 4.79	76.52 ± 5.49	0 ± 0	10.15 ± 4.79	17.50 ± 3.61	42.45 ± 10.27

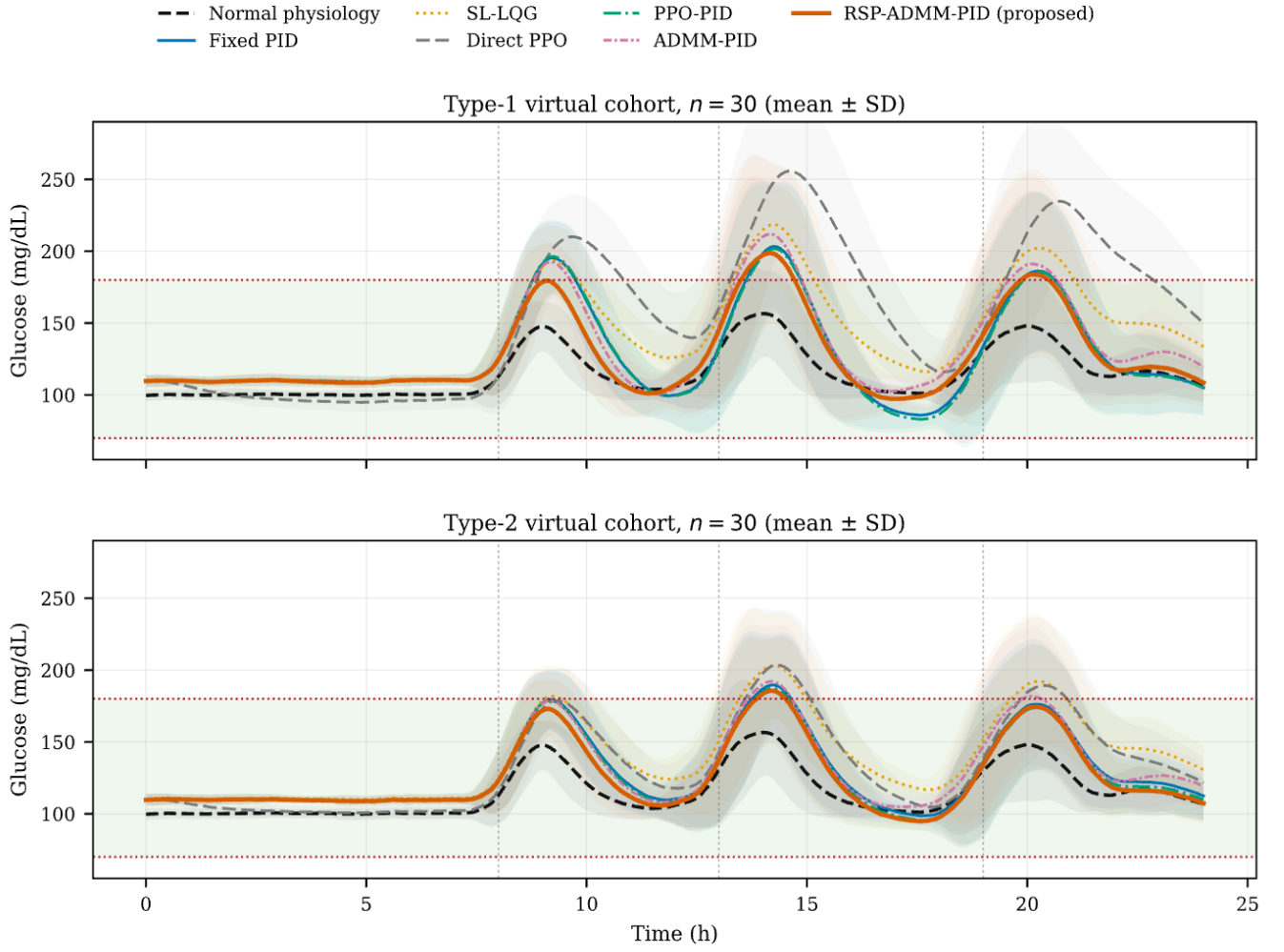


Figure 2. Held-out glucose trajectories (mean $\pm$ between-subject SD, $n = 30$) for *T1D* and *T2D*. Meal markers show the median seeded event times; individual meals vary in timing, size, and absorption. The green trace is the matched no-pump normal reference.

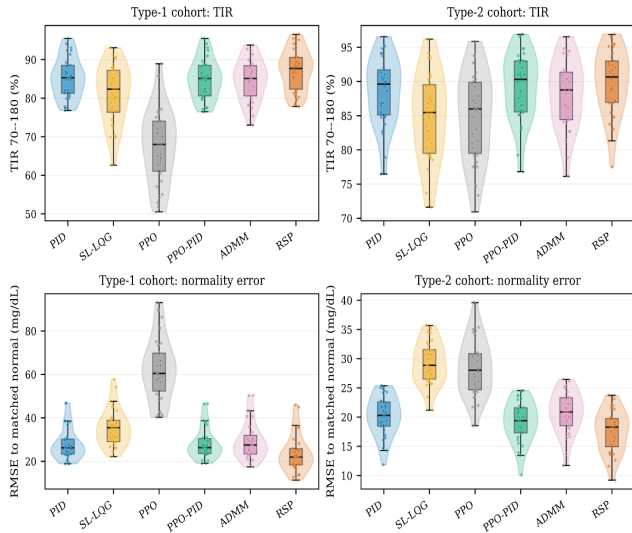


Figure 3. Per-subject outcome distributions. Violin density, box (quartiles), median and individual points are shown; each method uses identical subject and day seeds.

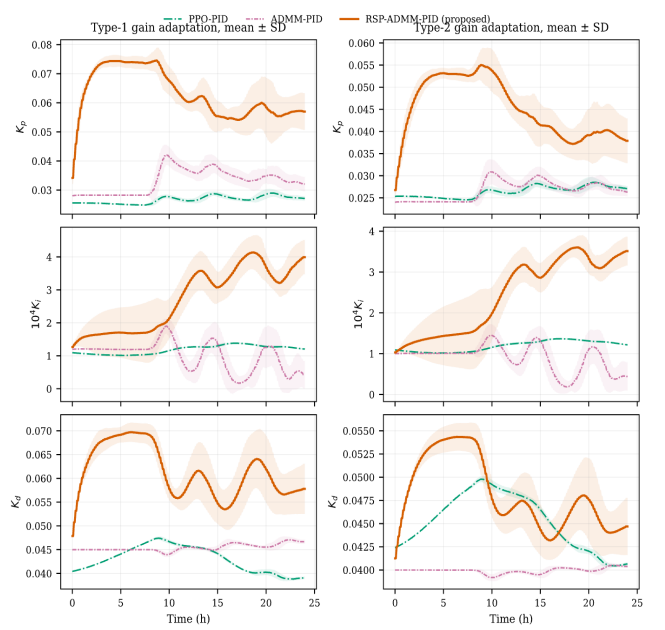


Figure 4. Adaptive gain trajectories for both held-out cohorts (mean $\pm$ SD, $n = 30$). All adaptive methods change K_p , K_i , and K_d ; prediction yields the most pronounced structured adaptation

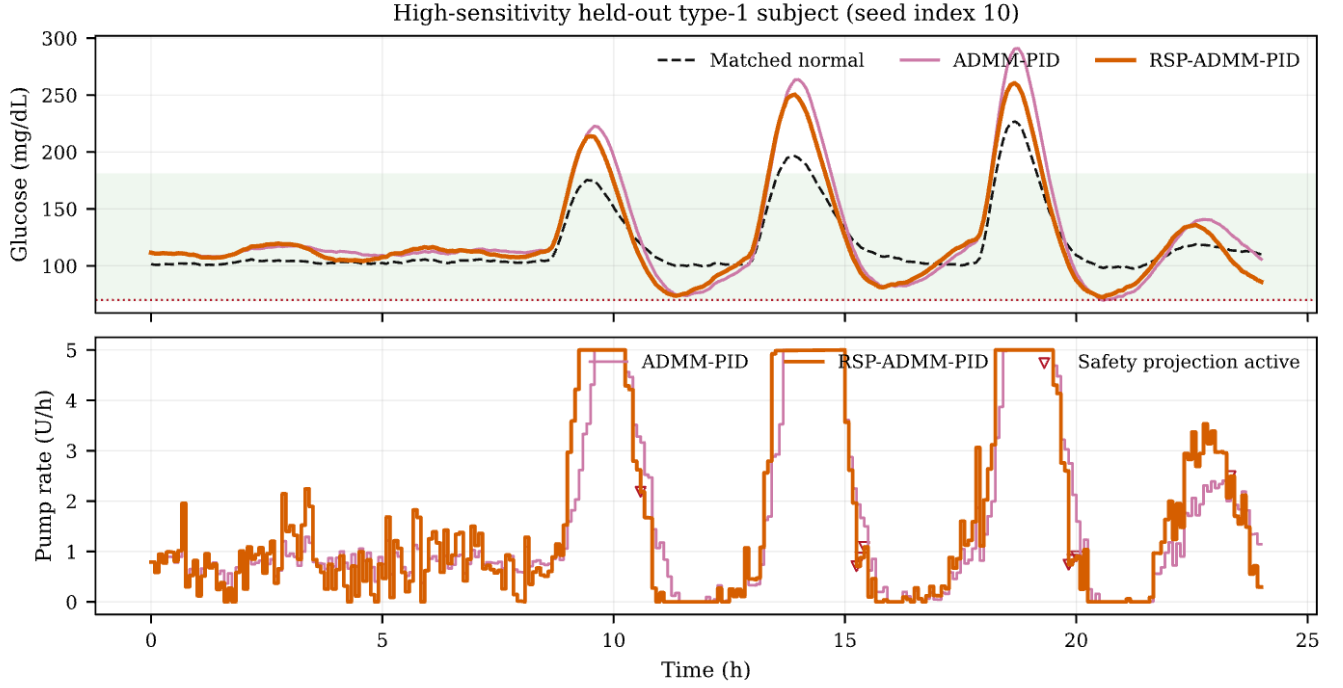


Figure 5. Projection markers identify commands reduced by Equation (Safety). The proposed tightened $75 \frac{\text{mg}}{\text{dL}}$ prediction floor removes the observed low sample while retaining post-meal correction.

Table 3. Effect of each component in proposed method (mean $\pm$ std, $n = \frac{15}{\text{cohort}}$). RMSE is to $110 \frac{\text{mg}}{\text{dL}}$; Int. is safety interventions per day.

	Variant	TIR	TBR	RMSE	Int.
T1D	ADMM	84.18 ± 5.09	.023	50.99 ± 11.15	0
	+scenarios	86.34 ± 5.46	.023	45.71 ± 11.21	0
	+safety	84.20 ± 5.10	0	50.98 ± 11.15	2.2
	full	86.44 ± 5.49	0	45.49 ± 11.36	17.4
T2D	ADMM	86.69 ± 5.35	0	42.91 ± 7.86	0
	+scenarios	88.35 ± 5.14	0	39.22 ± 7.36	0
	+safety	86.69 ± 5.35	0	42.91 ± 7.86	0
	full	88.35 ± 5.14	0	39.22 ± 7.36	0

D. Sensitivity analysis and gain dynamics

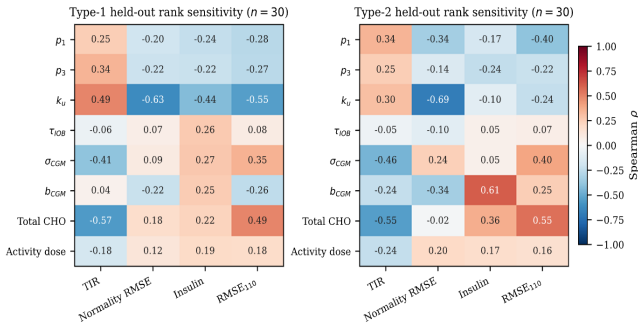


Figure 6. Exploratory held-out uncertainty sensitivity for the proposed method. Cells are Spearman rank correlations between sampled subject/day factors and outcomes; parameter ratios are relative to the cohort nominal value. CHO denotes total daily carbohydrate.

Figure 4 presents the dynamics of gain it ADMM, RSP-ADMM and PPO-PID. As seen in figure 4, RSP-ADMM has the most non-linear changes to compensate for the

optimization of gain. Moreover, PPO-PID and ADMM-PID, due to not having the safety filter, have smaller values for the gain. In addition, Figure 5 shows RSP-ADMM-PID and ADMM-PID difference in one held out subject, along with the difference in pump rate. The effect of the predictive filter can be seen clearly, as RSP-ADMM-PID applies the insulin sooner and causes smaller pick in glucose, making the trend more similar to a subject without diabetes.

Figure 6 depicts the sensitivity of the proposed method's performance metrics to variations in the physiological and disturbance parameters of (1). For T1D, the largest absolute Spearman rank correlation is 0.49, whereas for T2D it reaches 0.69. These values indicate that no single parameter completely determines the observed controller performance, although the T2D cohort exhibits a stronger dependence on particular physiological variations.

E. Effect of Kalman Filter and Predictive safety filter

Figure 7 illustrates the roles of the state-estimation and safety layers in the proposed architecture. In the upper panel, the CGM measurements exhibit the expected measurement variability, whereas the extended Kalman filter (EKF) estimate closely tracks the underlying glucose trajectory while suppressing high-frequency sensor fluctuations. Importantly, the estimator preserves the rapid glucose excursions following the seeded meals rather than excessively smoothing these control-relevant transients. This provides the scenario-based gain optimizer with a substantially clean estimate of the physiological state than the raw CGM signal alone.

The lower panel illustrates the action of the predictive safety projection in a representative case in which the nominal RSP-ADMM-PID command would lead to a predicted low-glucose excursion. Additionally, not having a safety filter, as demonstrated in Figure 7, would lead to hypoglycemia.

VI. DISSCUSSION AND CONCLUSION

The results show that for the proposed extended-Bergman benchmark, RSP-ADMM-PID improves glycemic regulation by combining short-horizon robust scenario optimization, online PID-gain adaptation and predictive safety projection. Across matched held-out T1D and insulin-treated T2D cohorts, the proposed method achieved the highest TIR of 75.83% on average and Lowest normality-RMSE (20.63 on average) against fixed PID, SL-LQG, direct PPO, PPO-PID and ADMM-PID, while producing no sampled glucose below 70 mg/dL. These improvements came with-in the approximately modeled dynamics, per real-value initialized 24-h simulation. The only trade-off of the proposed method was use of $20 \frac{U}{day}$ more insulin on average than lowest baseline.

These findings should nevertheless be interpreted as algorithmic evidence rather than physiological or clinical optimality. The model is needed to be calibrated to real patient records using the insulin pump. The 24-h experiments should be extended to longer-term adaptation, illness, stress, infusion failures, prolonged activity and correlated physiological variability.

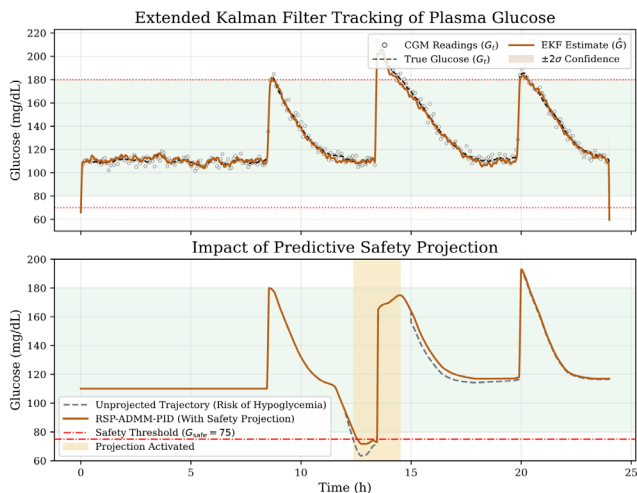


Figure 7. Performance of Kalman Filter in tracking blood Glucose, along with the impact of Safety Filter.

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